RIPHAH
INTERNATIONAL UNIVERSITY

BS-AI

LAB TASK 04

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Subject: Object Orientated Programming

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Task 1:

```java
package com.mycompany.lab4task1;

import java.util.Scanner;

public class Lab4task1 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Prime check
        System.out.print(s: "Enter a number to check prime: ");
        int primeNum = sc.nextInt();
        PrimeCheck.checkPrime(primeNum);

        // Array length
        System.out.print(s: "Enter size of array: ");
        int size = sc.nextInt();
        int[] arr = new int[size];
        System.out.println("Enter " + size + " elements:");
        for (int i = 0; i < size; i++) {
            arr[i] = sc.nextInt();
        }
        ArrayLength.findLength(arr);

        // Palindrome check
        System.out.print(s: "Enter a word to check palindrome: ");
        String word = sc.next();
        PalindromeCheck.checkPalindrome(word);
```

```java
        // Factorial
        System.out.print(s: "Enter a number to find factorial: ");
        int factNum = sc.nextInt();
        Factorial.findFactorial(factNum);

        // Swap using 3 variables
        System.out.print(s: "Enter two numbers to swap (3 variables): ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        SwapThree.swap(a, b);

        // Swap using 2 variables
        System.out.print(s: "Enter two numbers to swap (2 variables): ");
        int x = sc.nextInt();
        int y = sc.nextInt();
        SwapTwo.swap(x, y);

        sc.close();
    }
}
```

```java
package com.mycompany.lab4task1;

public class Factorial {
    public static void findFactorial(int num) {
        int fact = 1;
        for (int i = 1; i <= num; i++) {
            fact *= i;
        }
        System.out.println("Factorial of " + num + " = " + fact);
    }
}
```

```java
package com.mycompany.lab4task1;

public class PalindromeCheck {
    public static void checkPalindrome(String str) {
        String reversed = new StringBuilder(str).reverse().toString();
        if (str.equals(reversed)) {
            System.out.println(str + " is a Palindrome");
        } else {
            System.out.println(str + " is not a Palindrome");
        }
    }
}
```

```java
package com.mycompany.lab4task1;

public class PrimeCheck {
    public static void checkPrime(int num) {
        boolean isPrime = true;
        if (num <= 1) isPrime = false;
        else {
            for (int i = 2; i <= num / 2; i++) {
                if (num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }
        System.out.println(num + (isPrime ? " is Prime" : " is not Prime"));
    }
}
```

```java
package com.mycompany.lab4task1;

public class SwapThree {
    public static void swap(int a, int b) {
        System.out.println("Before swap: a=" + a + ", b=" + b);
        int temp = a;
        a = b;
        b = temp;
        System.out.println("After swap: a=" + a + ", b=" + b);
    }
}
```

```java
package com.mycompany.lab4task1;

public class SwapTwo {
    public static void swap(int a, int b) {
        System.out.println("Before swap: a=" + a + ", b=" + b);
        a = a + b;
        b = a - b;
        a = a - b;
        System.out.println("After swap: a=" + a + ", b=" + b);
    }
}
```

Task 2:

```
class Customer {
    String name;
    int customerId;
    void displayCustomer() {
        System.out.println("Customer Name: " + name);
        System.out.println("Customer ID: " + customerId);
    }
}
class BankAccount {
    int accountNumber;
    double balance;
    Customer customer;   // Association (has-a relationship)
    void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Deposited: " + amount);
    }
    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient Balance!");
```

```
        }
    }
    void displayAccount() {
        customer.displayCustomer();
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Current Balance: " + balance);
    }
    public static void main(String[] args) {
        Customer c1 = new Customer();
        c1.name = "Ahmed";
        c1.customerId = 5001;
        BankAccount acc1 = new BankAccount();
        acc1.accountNumber = 123456;
        acc1.balance = 10000;
        acc1.customer = c1;
        acc1.displayAccount();
        acc1.deposit(2000);
        acc1.withdraw(3000);
        acc1.displayAccount();
    }
}
```